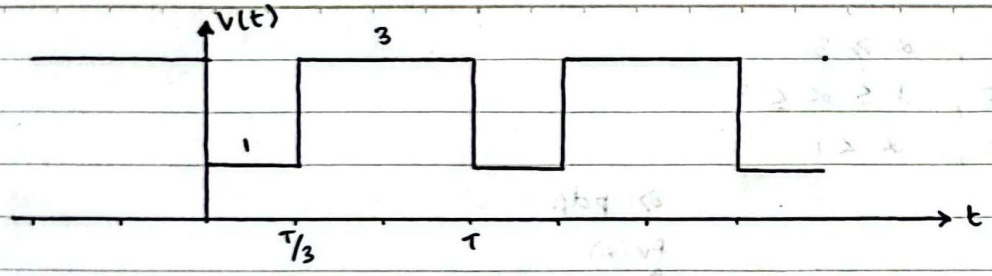
Elsadanika Maura Sitorus
211344013 - 2NK

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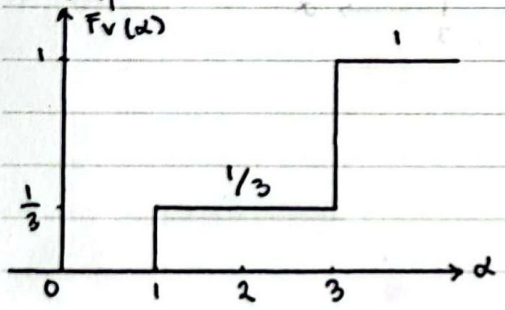
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Latihan Soal Probabilitas & Statistika

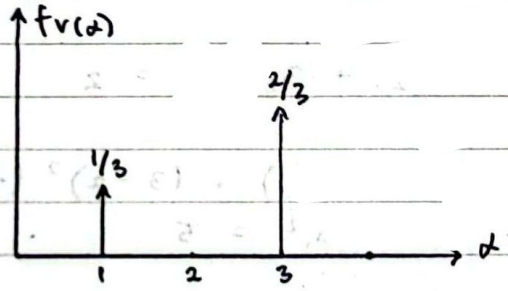
① 

$$F_V(\alpha) = P\{V \le \alpha\} = \begin{cases} 1, & 3 \le \alpha \\ 1/3, & 1 \le \alpha < 3 \\ 0, & \alpha < 1 \end{cases}$$

⇒ cdf



⇒ pdf



⇒ $m_1 = 1\left(\frac{1}{3}\right) + 3\left(\frac{2}{3}\right) = \frac{7}{3}$

$m_1^2 = \left(\frac{7}{3}\right)^2 = 49/9$

$\sigma^2 = (1 - 7/3)^2\left(\frac{1}{3}\right) + (3 - 7/3)^2\left(\frac{1}{3}\right)$

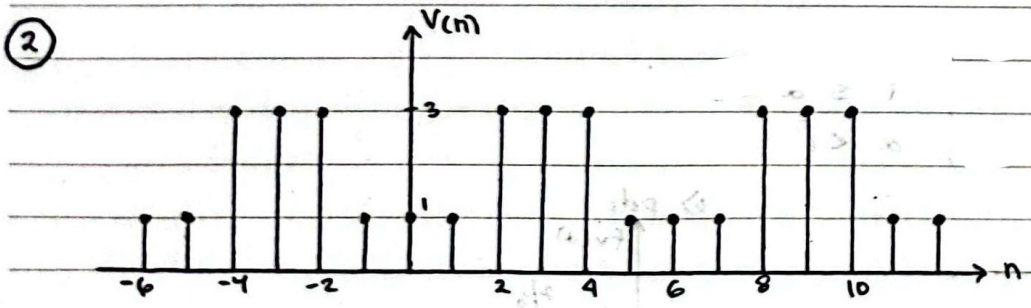
$= 16/27 + 4/27 = 20/27$

$m_2 = \sigma^2 + m_1^2 = 20/27 + 49/9 = 167/27$

⇒ $T = 5 \rightarrow P = \frac{1}{5}\int_0^5 \left[\frac{2}{3}(3)^2 + \frac{1}{3}(1)^2\right] dt$

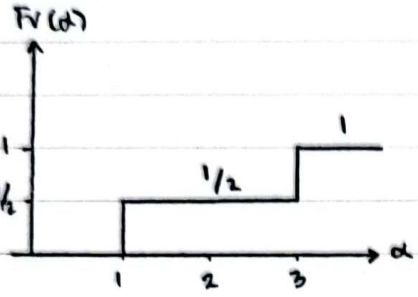
$P = \frac{1}{5}\int_0^5 \frac{19}{3}\, dt = \frac{19}{15} t\Big|_0^5 = \frac{95}{15}$

⇒ $E = \int_{-\infty}^{\infty} v^2(t)\, dt = \infty$

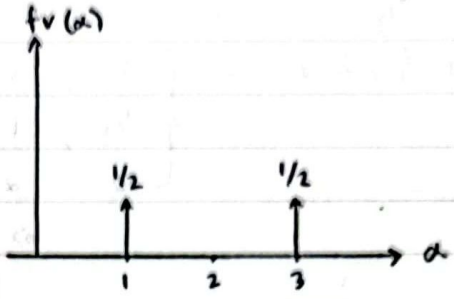
② 

$$F_V(\alpha) = \begin{cases} 1, & \alpha \geq 3 \\ \frac{1}{2}, & 1 \leq \alpha < 3 \\ 0, & \alpha < 1 \end{cases}$$

a) cdf



a) pdf



a) $m_1 = 1\left(\frac{1}{2}\right) + 3\left(\frac{1}{2}\right) = 2$

$m_1^2 = 4$

$\sigma^2 = (1-2)^2\left(\frac{1}{2}\right) + (3-2)^2\left(\frac{1}{2}\right) = 1$

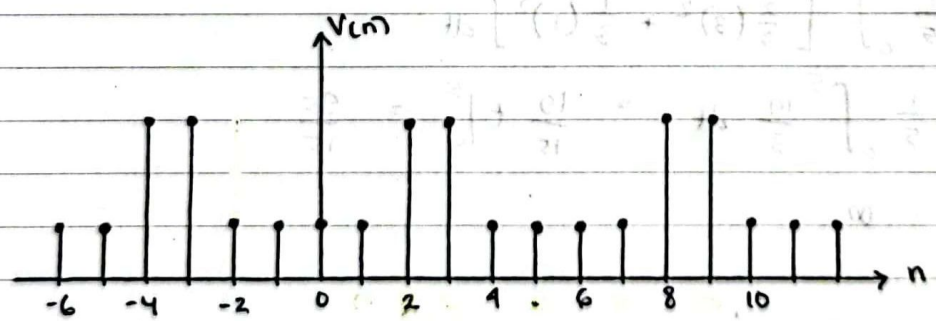
$m_2 = \sigma^2 + m_1^2 = 5$

a) $T = 5 \longrightarrow P = \frac{1}{5}\int_0^5 \left[\left(\frac{1}{2}\right)(1)^2 + \left(\frac{1}{2}\right)(3)^2\right] dt$

$= \frac{5}{5}\, t\Big|_0^5 = 5$

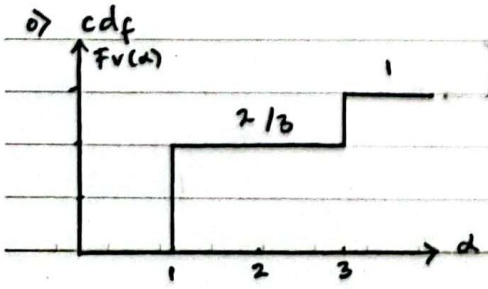
a) $E = \int_{-\infty}^{\infty} V^2(t)\, dt = \infty$

(3)

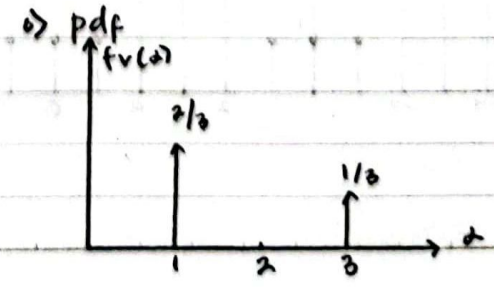


$$F_V(\alpha) = \begin{cases} 1, & \alpha \geq 3 \\ 2/3, & 1 \leq \alpha < 3 \\ 0, & \alpha < 1 \end{cases}$$

a) cdf



a) pdf



No.

Date

o) $m_1 = 1\left(\frac{2}{3}\right) + 3\left(\frac{1}{3}\right) = \frac{5}{3}$

$m_1^2 = 25/9$

$\sigma^2 = \left(1 - \frac{5}{3}\right)^2\left(\frac{2}{3}\right) + \left(3 - \frac{5}{3}\right)^2\left(\frac{1}{3}\right) = 8/9$

$m_2 = \sigma^2 + m_1^2 = 25/9 + 8/9 = 33/9$

o) $T = 5 \longrightarrow P = \frac{1}{5}\int_0^5 \left[\frac{2}{3}(1)^2 + \frac{1}{3}(3)^2\right] dt$

$= \frac{1}{5} \cdot \frac{11}{3} \cdot t\Big|_0^5 = \frac{1}{3}$